ASSIGNMENT 5 — QUESTION 6

Person to Faculty to Professor

JavaScript (ES6+) solution

```
function Person(name) {
    this.name = name;
}
Person.prototype.introduce = function () {
    console.log(`Person: ${this.name}`);
};

function Faculty(name, department) {
    Person.call(this, name);
    this.department = department;
}
Faculty.prototype = Object.create(Person.prototype);
Faculty.prototype.constructor = Faculty;
Faculty.prototype.showDepartment = function () {
    console.log(`Department: ${this.department}`);
};

function Professor(name, department, subject) {
    Faculty.call(this, name, department);
    this.subject = subject;
}
Professor.prototype = Object.create(Faculty.prototype);
Professor.prototype.constructor = Professor;
Professor.prototype.showSubject = function () {
    console.log(`Subject: ${this.subject}`);
};

const professor = new Professor("Dr. Sharma", "CSE", "JavaScript");
professor.introduce();
professor.showDepartment();
professor.showSubject();
```

Expected Output

```
Person: Dr. Sharma
Department: CSE
Subject: JavaScript
```

Explanation

The chain is professor → Professor.prototype → Faculty.prototype → Person.prototype. A Professor instance can access its own method and methods inherited from Faculty and Person.